P1-08 — Integration Audit (Blueprint Correction Freeze)

Command: P1-08 — Apply Commander Decisions and Freeze the Corrected Blueprint **Date:** 21 October 2026 **Decisions applied (explicit Commander decisions, not re-decided here):** PC-1 APPROVE, PC-2 APPROVE, PC-3 APPROVE, PC-4 APPROVE, PC-5 APPROVE, PC-6 REJECT (obsolete “to be proved” label; proved C6 status/formulas recorded instead).

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1. Pre-edit state (recorded before any edit)

- Branch: arena/ab8e330d-galios-hull; working tree clean.
- Local HEAD before editing: a92bb21a29f43a782bc973eff34203aaab8e0aae (P1-07). Remote branch ref (via git ls-remote): a92bb21a29f43a782bc973eff34203aaab8e0aae — **local = remote, no divergence**, so the controlled edit proceeded (no reset, no overwrite; none was needed or performed).

2. Blueprint hashes

- **Before:** SHA-256 013ef5a20742b3de735d49edb6f2fdde074b0d134adea9c85d30d24cab739ce3
- **After:** SHA-256 ec2fd8d58f8d2a986314f83997d750d689f7c88e98ec8f87753ab4c9cf438cff
- **New frozen state:** blueprint at the after-hash, contained in integration commit **1c67fda** (“P1-08: apply PC-1..PC-5, reject PC-6 label; re-freeze blueprint”), pushed to origin/arena/ab8e330d-galios-hull and verified against the remote ref after push (see §6; final remote verification is reported in the P1-08 final report, together with this audit’s own commit hash).

3. Exact accepted changes and locations (one controlled edit; nothing else)

All line numbers refer to the after-edit file. Each change carries a dated [P1-08 ...] marker in the file, and a “Correction Notes” section (lines 6–8) summarizing all six decisions.

| Decision | Location | Change | Proof reference

1 | Header/freeze record | Lines 2, 6–8 | Status line updated; dated Correction Notes section added listing PC-1…PC-5 applied and PC-6 rejected, with rationale | This audit; P1_07_PC_DECISION_DOSSIER.md

2 | PC-2 (dual order) | §0, line 17 | Added D(C) = {y : ⟨c, y⟩_{s,k} = 0 for all c ∈ C} (code first, candidate second); stated D(C) = ρ(C^⊥), Euclidean generator h^↯ (ordinary reciprocal), code-first generator h_a# = ρ̃(h_a^↯) | P1_04_PROOFS_T1_T4.md (S7), Lemma RR-3, Prop. 4.2, T3

3 | PC-3 (# domain) | §0, line 24; T2(i), line 38 | # restricted to monic f with f(0) ≠ 0; noted automatic for `f | Ṁ_s (Ṁ_s(0) = −μ_s ≠ 0) and h_a` | P1_04_PROOFS_T1_T4.md (S6), Lemma RR-2, §7 audit

4 | PC-5 (T1 scope) | T1, line 34 | Global code = A-submodule of A^n under the fixed labeled CRT decomposition; component codes K_s-linear; counts labeled, not equivalence classes; product parametrization not claimed for mixing F_q-subspaces | P1_05_PROOFS_T5_T6_COROLLARIES.md (G3), Prop. 3.1

5 | PC-4 (C4 count) | C4, line 84 | Added #{a : D(C) ⊆ C} = ∏_O tr((B′_P)^{L_O}), (B′_P)_{a,b} = 1 iff a + b ≤ P, under the characterization + compatible/labeled hypotheses; noted equality of the two counts via α_i ↦ P − α_i | P1_05_PROOFS_T5_T6_COROLLARIES.md §5 (C4)

6 | PC-1 (C5 mean) | C5, lines 85–87 | Hull mean replaced by E[Hull] = μ_P·Σ_{O : L_O ≥ 2} w_O L_O + ν_P·Σ_{O : L_O = 1} w_O (μ_P = P(2P+1)/(6(P+1)), ν_P = ⌊P²/4⌋/(P+1)), old display explicitly marked incorrect for L_O = 1; proof key corrected (distinct uniforms only for L_O ≥ 2; self-loop min(P−A, A) for L_O = 1); **P = 1 case corrected**: E[Hull] = (1/4)·Σ_{L_O ≥ 2} w_O L_O since ν_1 = 0. Code-dimension mean E[dim C] = (n/2)·Σ_s m_s **preserved unchanged** | P1_06_MOMENT_PROOFS_C5_C6.md §2 (Lemmas 1.1, Props. 2.2–2.3, Theorem 2.4); P1_03_MATHEMATICAL_PROOF_PLAN.md §4

7 | PC-6 (rejected label; correct status) | C6, lines 88–92 | **No** “to be proved” label added. C6 now states the proved result: Var(dim_{F_q} Hull_k) = Σ_s Σ_{O∈O_s} w_O²·V_{L_O}(P) with V_1 parity-split, V_2, V_L (L≥3) formulas, adjacent covariance κ_P, L=2 cross-covariance; mean-vs-variance weights (w_O vs w_O²) distinguished; marked “RR-8 discharged” | P1_06_MOMENT_PROOFS_C5_C6.md §3 (Theorem 3.7)

Diff audit: the complete git diff of the blueprint was inspected line by line; it contains exactly the seven items above (status/correction-notes, §0 ×2, T1, T2, C4, C5, C6). No other mathematical

change was introduced; in particular T1–T6 statements, C1–C3, C7, the layer-decomposition section, the dependency map, and the section architecture are byte-unchanged.

4. Consistency audit (against the proof files)

- **C5 at P = 1 / L = 1:** blueprint now gives ν_1 = ⌊1/4⌋/2 = 0 and E[Hull] = (1/4) Σ_{L_O≥2} w_O L_O, matching P1-06 Prop. 2.2/Theorem 2.4 and the brute counterexamples to the old display (GF(2)/GF(8): 11/3 vs old 35/9; GF(3)/GF(9) λ=1: 2 vs 7/2).
- **Dual convention:** the added sentence is verbatim the convention (S7) of P1-04 under which h_a# was proved to be the dual generator; h^♮ (ordinary reciprocal, RR-3) vs h_a# = ρ̃(h_a^♮) (RR-4 transport) stated as in Prop. 4.2; the opposite-order alternative (which would give the σ-reciprocal) is excluded by the sentence, as PC-2 intended.
- **Reciprocal domain:** f(0) ≠ 0 in §0 and T2 matches P1-04 (S6)/RR-2; necessity witness f(x) = x recorded in the dossier.
- **Dual-containing count:** formula, matrix B′_P, and hypotheses (characterization + compatible twists + labeled model) match P1-05 §5 exactly; count equality via α ↦ P − α noted.
- **C6 variance:** formulas compared character-for-character with P1-06 Theorem 3.7 (V_1 even $P(P+2)(P^2+2P+4)/(48(P+1)^2)$, odd $(P-1)(P+3)/48$; V_2 = $P(P+2)(P^2+2P+3)/(18(P+1)^2)$; V_L = $L\cdot P(P+2)(P^2+2P+2)/(60(P+1)^2)$; κ_P = $-P(P+2)(7P^2+14P+9)/(360(P+1)^2)$; L=2 covariance $-P^2(P+2)^2/(36(P+1)^2)$) — identical. Weights: mean uses w_O (P1-06 Thm 2.4), variance w_O^2 (Thm 3.7) — distinguished in both C5/C6 text now.
- **Compatibility assumptions:** unchanged everywhere (per-component λ_s^{1+p^{d_s−k}} = 1); PC-4's count explicitly inherits them.

5. Validation (existing scripts, unmodified — supporting evidence, not proof)

Run on 21 Oct 2026 after the edit:

```
python3 code/validate_p1_03.py        # exit 0; output byte-identical to code/validation_p1_03_output.txt;
ALL P1-03 EXTENDED CHECKS PASSED
python3 code/validate_p1_05.py        # ALL P1-05 MULTI-COMPONENT ASSEMBLY CHECKS PASSED
python3 code/validate_p1_06.py        # ALL P1-06 MOMENT CHECKS PASSED
python3 code/validate_repeated_root.py # ALL PROTOTYPE CHECKS PASSED
```

No validation script or saved output was modified (git status shows only the intended files). Passing finite tests support — but do not substitute for — the cited proofs.

6. Freeze, remaining gaps

- **Frozen state identifier:** blueprint SHA-256 (after) ec2fd8d58f8d2a986314f83997d750d689f7c88e98ec8f87753ab4c9cf438cff, committed as 1c67fda and pushed; the remote ref was re-read after push to confirm it points at the pushed history (no force-push, no history rewrite at any point).
- **Closed by this integration:** GAP-01 (C5 mean), GAP-06/RR-8 (variance), GAP-17 (convention/corrections register). PC-1…PC-5 are now **APPROVED–APPLIED (21 Oct 2026)**; PC-6 is **REJECTED (superseded)** with the proved C6 status recorded.
- **Still open (unchanged): Gate G1 — novelty/literature** (repo L1 theorem text unresolved; no novelty or priority claim is made anywhere in the corrected blueprint); C7/RR-9 remains a conditional scope statement (no EAQECC parameters from the enumerator alone); incompatible twists remain out of scope; counts are labeled assignments only. Manuscript drafting has **not** begun.